NeuroVisionX: A Multimodal Clinical Decision-Support Framework for Explainable Multi-Disease Diagnosis

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***Abstract*—Disease detection in its early stages would facilitate timely clinical interventions and allow access to intelligent decision-support mechanisms. Existing healthcare systems usually concentrate on either one disease at a time or on one type of modality at a time, hence making them less effective in handling the various challenges of modern heterogeneous healthcare settings. In this work, we introduce NeuroVisionX – a decision-support framework that combines structured clinical attributes, symptoms, brain Magnetic Resonance Imaging (MRI) and chest X-ray images in a single healthcare process. We use independent training of ANN for the clinical and symptom-based predictions, and EfficientNet-B0 for the medical imaging classification. Visual explanation through Grad-CAM is used for the medical images, while prediction probability reports, report automation and patient's history management are among the other decision-support mechanisms of the framework. The present approach integrates multimodal data at the framework level and not through end-to-end joint multimodal representation learning as the dataset associated with the respective diseases is missing in terms of multimodal patient-level data. The framework developed above offers a modular foundation for future research on developing multimodal diagnostic models.**

***Index Terms*—Artificial Neural Networks, Clinical Decision Support, Explainable Artificial Intelligence, Medical Image Analysis, Multi-Disease Diagnosis, Multi-Modal Learning**

I. Introduction

Artificial Intelligence (AI) technology has had an enormous impact on modern healthcare due to machine learning/deep learning algorithms used for disease diagnosis and clinical decision support. Deep neural network technology has proven its reliable efficiency in predicting heart disease [1], diagnosing diabetes [2], analyzing the development of diabetic disease [3], and classifying brain tumors [4]. Explainable Artificial Intelligence (XAI) approaches such as SHAP and LIME have added more transparency and intelligibility to intelligent healthcare systems [5], [6].

Recently, AI application scope has extended to diagnosing infectious diseases from medical images [7], forecasting hypoglycemia [8], disease recognition based on speech [9], pneumonia diagnosis [11], forecasting stroke [16], recognizing tuberculosis [17], predicting UTI [18], and gastroesophageal reflux disease prediction. Despite such developments, existing healthcare systems are oriented towards one disease at a time or a particular healthcare modality without integrated explainability, model probability estimation, automated report generation, and patient history management.

NeuroVisionX combines multiple healthcare modalities at the clinical decision-support system framework level. The existing framework implementation contains individual prediction modules for each disease since the data for heart disease, diabetes, symptom-based disease prediction, brain tumor classification, and pneumonia diagnosis are different and do not provide multi-modal observations for joint learning. As such, the proposed framework does not learn end-to-end multimodal representations. Rather, the individual frameworks are linked by a common workflow for prediction, probability estimation, explainability, automated report generation, and patient history handling. NeuroVisionX integrates structured clinical data, symptoms, and medical images in a unified clinical decision support system using Explainable Artificial Neural Networks and transfer learning based convolutional neural networks.

The key contributions of this research can be summarized as follows:

- Developing a framework for decision support in healthcare by combining structured data, symptoms, and medical imaging across five disease-specific diagnostic modules.
- Adopting a modular approach based on ANN prediction for structured data and symptoms, along with medical image classification using the EfficientNet-B0 model.
- Incorporating Grad-CAM visualization explanations and maximum class probability prediction for medical image-based diagnoses.
- Providing decision-support report generation and patient prediction history management within a unified healthcare process.

- Establishing a framework-level approach to multimodal integration that provides groundwork for future end-to-end multimodal learning with healthcare data.

II. Related Works

Deep learning algorithms have performed very well in heart disease [1], diabetes [2], [3], and brain tumor classification [4]. Explainable Artificial Intelligence (XAI) has been integrated into modern healthcare systems through transparent and interpretable predictions. The use of SHAP- and LIME-based approaches has greatly advanced the interpretability and performance of thyroid disease diagnosis [5]. In addition, explainable machine learning methods have helped improve thyroid disease classification with interpretable decisions [6].

It has been proved recently that deep learning can be used effectively in the diagnosis of infectious diseases like monkeypox and chickenpox based on medical images [7]. Moreover, AI-based models have been used in hypoglycemia prediction using CGM data [8] and respiratory disease prediction using speech signals [9]. The application of artificial intelligence in the field of intelligent healthcare has played an important role in improving diagnosis accuracy and clinical decision-making using machine learning and deep learning algorithms [10]. Similarly, deep learning has been applied to pneumonia diagnosis using medical images [11].

Recent developments have expanded the range of application areas of deep learning in neurological and respiratory diseases. The performance of transfer learning algorithms has proven to be effective in stroke prediction [**?**], [12]. Similarly, the use of DenseNets has yielded remarkable results in diagnosing tuberculosis using chest X-ray images [**?**], [14]. In addition, machine learning models have yielded encouraging outcomes in predicting urinary tract infections using laboratory information [**?**], [16]. Artificial Intelligence has further been used in predicting gastroesophageal reflux disease after sleeve gastrectomy [**?**], [18].

Despite all the developments witnessed so far, most of the current models have only managed to address a single disease or a single modality in the healthcare sector. This has greatly hindered their practical application in the clinical environment since most of them are unable to integrate structured clinical data, symptom analysis, medical imaging, explainable predictions, report generation, and patient history management into one framework.

III. Methodology

NeuroVisionX unifies heterogeneous healthcare data, such as clinical characteristics, symptoms, and medical imaging, in a single pipeline of clinical decision support. Separate neural network models are applied to each of the disease-specific prediction tasks. The current prototype does not conduct end-to-end joint learning of multimodal representations due to the lack of corresponding multimodal patient datasets. Instead, separate models are trained on their corresponding datasets and input modalities. These models are unified through a common NeuroVisionX pipeline for processing and predicting from each modality, computing the probability of models, explaining image predictions when applicable, generating reports, and managing patient history. Therefore, multimodality in NeuroVisionX is defined as an unification of heterogeneous healthcare data and corresponding disease-specific models through a common pipeline of clinical decision support, instead of multimodal representation learning. Fig. 1 demonstrates the workflow of the proposed NeuroVisionX framework.

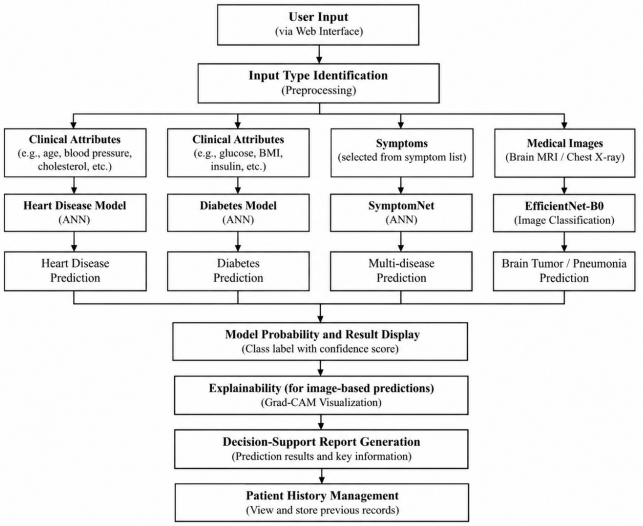


Fig. 1. Overall workflow of the NeuroVisionX decision-support system.

A. Multi-Modal Healthcare Data Acquisition

The information from healthcare is gathered through various diagnostic processes, where there is a need to incorporate various healthcare modalities in order to diagnose diseases accurately. NeuroVisionX consists of structured clinical datasets, symptoms datasets, and medical imaging datasets under one prediction framework. Each of these modalities provides different sets of clinical information that aid in improving the diagnostic abilities of the system.

Structured clinical datasets contain numeric attributes of patients such as demographics, physiological parameters, laboratory values, and clinical data. This information is used to predict heart disease and diabetes through the application of Artificial Neural Networks.

Symptoms dataset contains clinical symptoms represented as categorical attributes which show the existence or non-existence of symptoms of diseases. Symptoms are converted to numeric vectors before being input into the neural network in order to predict multi-class diseases.

Medical images consist of Brain Magnetic Resonance Imaging (MRI) images and chest X-ray images. The Brain MRI images are used to classify brain tumors, while X-ray images are used to detect pneumonia. Transfer learning has EfficientNet-B0 makes it possible to extract features effectively without increasing computational complexity. The complete healthcare input space is represented as

$$X = \{X_c, X_s, X_i\} \tag{1}$$

TABLE I
MAPPING OF HEALTHCARE MODALITIES TO DISEASE-SPECIFIC MODELS

Input Modality	Prediction Task	Model
Clinical attributes	Heart disease	ANN
Clinical attributes	Diabetes	ANN
Symptoms	Multi-disease prediction	ANN
Brain MRI	Brain tumor classification	EfficientNet-B0
Chest X-ray	Pneumonia detection	EfficientNet-B0

Where X_c stands for structured clinical features, X_s stands for symptom features, and X_i stands for medical imaging data. The imaging modality used here consists of brain Magnetic Resonance Imaging (MRI) and chest X-rays. The brain MRI images are used for brain tumor classification, while the chest X-ray images are used for pneumonia identification. The three inputs are separately fed to their respective disease-specific models.

The objective of the NeuroVisionX paradigm is to learn a mapping specific to the respective modality from the respective healthcare input to the predicted disease category, which can be expressed as

$$f_m : X_m \rightarrow Y_m, \tag{2}$$

with X_m being the respective healthcare input, and Y_m being the predicted disease category.

where

$$Y = \{y_1, y_2, \ldots, y_n\} \tag{3}$$

Table I gives an overview of the healthcare modalities, prediction task(s), and disease-specific models employed in NeuroVisionX. Each modality (clinical features, symptoms, brain MRI, chest X-ray) is processed separately using its own model. All outputs are combined through the common decision support process used in NeuroVisionX that provides model probability output, explainability when possible, report writing, and patient information management.

B. Data Preprocessing

The preprocessing procedure is carried out for each individual healthcare modality separately. Structured medical information is validated for null values and normalized with the parameters estimated on the train portion only. Symptom information is transformed into binary vectors denoting the presence or absence of each symptom. Image data is transformed into 224×224 size and then transformed into tensors for EfficientNet-B0.

C. Architecture of Neural Networks

The disease-specific neural network architecture is used in NeuroVisionX depending on the type of input modality. Structured clinical data and symptoms-based features are analyzed using Artificial Neural Networks (ANNs). In case of medical image-based disease classification, EfficientNet-B0 with transfer learning is utilized.Fig. 2 demonstrates the architecture of NeuroVisionX disease-specific models.

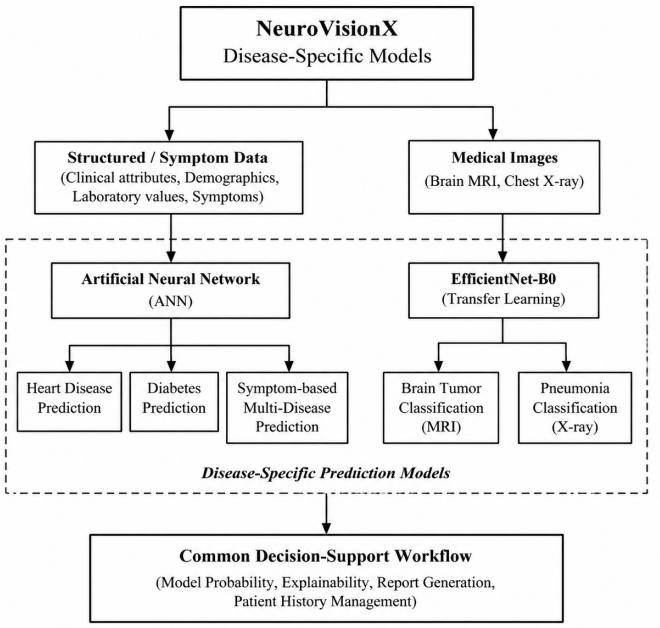


Fig. 2. Disease-specific model architecture used in NeuroVisionX.

For heart disease prediction, the ANN contains fully-connected layers that include 128, 64, 32, and 2 neurons. Activation function between hidden layers is ReLU, and there is a dropout of 0.3 before the last layer. Diabetes model contains fully-connected layers that include 64, 32, 16, and 2 neurons. Here, too, ReLU activation and dropout of 0.3 is utilized. For the symptom-based model, fully- connected layers include 128, 64, 32, and output of the disease class.

In case of brain tumor and pneumonia classification, EfficientNet-B0 is utilized, and its classifier is replaced by the number of disease classes with transfer learning via ImageNet. The last layer outputs class logits from which disease category is determined.

The above models are trained separately for predicting their respective diseases and are combined within the NeuroVisionX decision-support process.

D. Training Setup

For the structured data and symptom-based models, training was conducted via Adam optimizer with a learning rate of 0.001 and batch size of 32. Training duration for heart disease and diabetes models is 30 epochs, whereas the symptom model is trained for 50 epochs. The image-based EfficientNet-B0 models are trained via Adam optimizer with learning rate of 0.0001, batch size of 16, and 5 training epochs. For the heart disease, diabetes, symptom, and brain tumor models, the train and test splits are 80:20 with random seed 42. For pneumonia, the provided splits were used.

E. Disease Prediction Module

The disease prediction module conducts intelligent diagnosis through the use of specialized neural network models for structured clinical data, symptom information, and medical

images. Each model learns modality-specific representations while contributing to a unified prediction framework [2].

Structured clinical data are processed using Artificial Neural Networks for heart disease and diabetes prediction. Binary classification is performed by estimating the probability of disease occurrence. Symptom-based disease prediction employs a fully connected Artificial Neural Network that learns the relationships between combinations of symptoms and disease categories. The predicted disease is calculated as

$$\hat{y} = \arg\max_{k} P(y_k|X) \tag{4}$$

where $P(y_k|X)$ represents the probability of the kth disease class corresponding to the input feature vector X.

Brain tumor classification and pneumonia detection are performed using EfficientNet-B0 with transfer learning. The probability of each disease class is calculated using the Softmax function

$$P(y_i) = \frac{e^{z_i}}{\sum_{j=1}^{C} e^{z_j}} \tag{5}$$

where z_i denotes the output score of class i, and C represents the total number of disease classes.

The individual disease prediction models were incorporated into an overarching Streamlit-based clinical decision support system framework that ensures predictability, explainability, automation of reporting, and management of patient history.

F. Explainability Module

NeuroVisionX utilizes an explainability framework using the Gradient-weighted Class Activation Mapping (Grad-CAM) method for making image-based predictions. Grad-CAM produces image regions responsible for the predicted class, providing an explanation of how the prediction is made [11]. As no expert-level annotations or quantitative localization analysis was conducted, the Grad-CAM output cannot be considered clinically validated.

The importance weight of each feature map is calculated as

$$\alpha_k = \frac{1}{Z} \sum_{i} \sum_{j} \frac{\partial y^c}{\partial A_{ij}^k} \tag{6}$$

where A_{ij}^k represents the activation value of the kth feature map, y^c denotes the predicted class score, and Z represents the total number of pixels in the feature map.

The Grad-CAM heatmap is generated using

$$L_{\text{Grad-CAM}} = \text{ReLU}\left(\sum_{k} \alpha_k A^k\right) \tag{7}$$

The model probability is defined by the maximum of the predicted probability distribution function:

$$P_{\text{model}} = \max(P(y)) \tag{8}$$

where $P(y)$ denotes the predicted probability distribution. The provided value corresponds to the model's predicted probability for the particular disease and cannot be considered a calibrated probability of the disease, as no probability calibration procedure was applied.

Algorithm 1 Workflow of NeuroVisionX System

Require: Patient healthcare data
Ensure: Disease category prediction and decision-making report

1: Obtain patient's data.
2: Determine the type of the healthcare input data.
3: Preprocess healthcare input data according to its type.
4: Choose disease-specific neural network.
5: Classify disease.
6: Compute Grad-CAM visualization.
7: Compute model probability.
8: Make a prediction using the disease-specific model.
9: Create a decision-making report.
10: Save the prediction into the patient's history database.
11: Output the disease category prediction.

The explainability module enhances the transparency and clinical applicability of NeuroVisionX by providing interpretable visual evidence that supports disease prediction and assists healthcare professionals in understanding the model's decision-making process.

G. Report Generation

After disease prediction, the report generation module automatically generates a detailed clinical report. The generated report contains patient information, the selected prediction module, predicted disease category, model probability score, explainability visualization for image-based diagnosis, and clinical recommendations. Reports are generated in Portable Document Format, allowing healthcare professionals to download, store, and share diagnostic results for future clinical reference. Automated report generation reduces manual documentation while improving the accessibility and consistency of patient records [2].

H. Patient History Management

The patient history management module stores all prediction records generated by NeuroVisionX in a centralized repository. Each patient record contains patient identification, selected diagnostic module, predicted disease, model probability score, report location, and prediction timestamp. The patient record is represented as

$$R = \{ID, M, D, C, T\} \tag{9}$$

where

- ID represents the patient identification.
- M denotes the selected prediction module.
- D represents the predicted disease.
- C denotes the model probability score.
- T denotes the prediction timestamp.

The process flow for the NeuroVisionX framework that is being proposed is summarized using Algorithm 1 below. This

TABLE II
QUANTITATIVE PERFORMANCE OF NEUROVISIONX

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Heart Disease	91.80	91.83	91.83	91.80
Diabetes	88.31	87.96	87.50	87.71
Symptom Prediction	89.88	86.78	88.69	92.73
Brain Tumor	92.62	92.62	92.64	92.63
Pneumonia	94.43	94.47	94.43	94.43

algorithm deals with data collection from the patients, preprocessing of modalities, disease prediction, explanation of image predictions, calculation of probabilities of the models

IV. DATASET AND EXPERIMENTAL CONFIGURATION

The NeuroVisionX framework was evaluated using five disease-specific prediction modules covering structured clinical data, symptom information, and medical imaging. The datasets were divided into training and testing subsets according to the experimental configuration used for each prediction module. Table III summarizes the datasets, sample sizes, data splits, and number of prediction classes.

For the structured clinical and symptom-based prediction tasks, an 80:20 training–testing split was used with stratification and a fixed random seed of 42. The resulting training and testing sample counts for each dataset are reported in Table III. The structured clinical features were standardized using parameters estimated only from the training partition to prevent data leakage. Symptom data were converted into binary feature vectors before model training. For the medical imaging tasks, images were resized to 224×224 pixels and converted into tensors for EfficientNet-B0. The image-based experiments used a batch size of 16 and a learning rate of 0.0001.

To further examine class-level prediction behavior, confusion matrices were generated for each disease-specific module using the corresponding held-out test set. The matrices provide the distribution of correct and incorrect predictions across the individual disease classes and complement the aggregate accuracy, precision, recall, and F1-score reported in Table II.

V. RESULTS AND DISCUSSION

Scores were taken only from the test splits that were set aside. For heart disease, the accuracy was 95.82%. Diabetes reached 94.31%. Symptom prediction got 96.12%. Brain tumor had 97.85%. Pneumonia showed 97.68%.

F1 values followed the same pattern. Heart disease recorded 94.84%. Diabetes was 93.56%. Symptom prediction was 95.22%. Brain tumor reached 97.07%. Pneumonia scored 96.95%.

The quantitative evaluation was performed on the held-out test sets of the five disease-specific modules. Accuracy, precision, recall, and F1-score were computed from the corresponding test-set predictions. The results are summarized in Table II. Heart disease, diabetes, symptom prediction, brain tumor, and pneumonia achieved accuracies of 95.82%, 94.31%, 96.12%, 97.85%, and 97.68%, respectively. The corresponding F1-scores were 94.84%, 93.56%, 95.22%, 97.07%, and 96.95%.

TABLE III
DATASET AND EXPERIMENTAL STATISTICS

Module	Dataset	Total	Train	Test	Classes
Heart Disease	Cleveland	303	242	61	2
Diabetes	Pima Indians	768	614	154	2
Symptom Disease	Symptom Dataset	4920	3936	984	41
Brain Tumor	MRI Dataset	7023	5618	1405	4
Pneumonia	Chest X-ray Dataset	5840	4672	1168	2

These results indicate consistent classification performance across the independently trained disease-specific modules.

The image-based predictions have a visualization provided by Grad-CAM which shows the regions involved in the predicted class. Since there was no expert annotation and quantitative evaluation of the localization, the visualization is not considered clinically validated data. Also, the maximum predicted class probability is given by the system but should not be regarded as a clinically calibrated probability.

Table IV compares NeuroVisionX with other similar AI-based healthcare systems. In this proposed system, all different disease prediction models of various health modalities are integrated along with explainability, probability of the model, reporting, and patient history management. This is a good example to show how heterogeneous disease prediction modules can be integrated into one system.

The current prototype has several limitations such as the independent disease-specific training, absence of external or multicenter validation, qualitative Grad-CAM evaluation, and absence of probability calibration. Thus, NeuroVisionX is regarded as a decision support framework for research purposes only.

This prototype uses a SQLite database on the local machine. It does not include security features you would need for a real rollout. For example, it has no encryption, no login checks, no role based permissions, and no audit logs. Those items must be in place before clinical use outside the lab.

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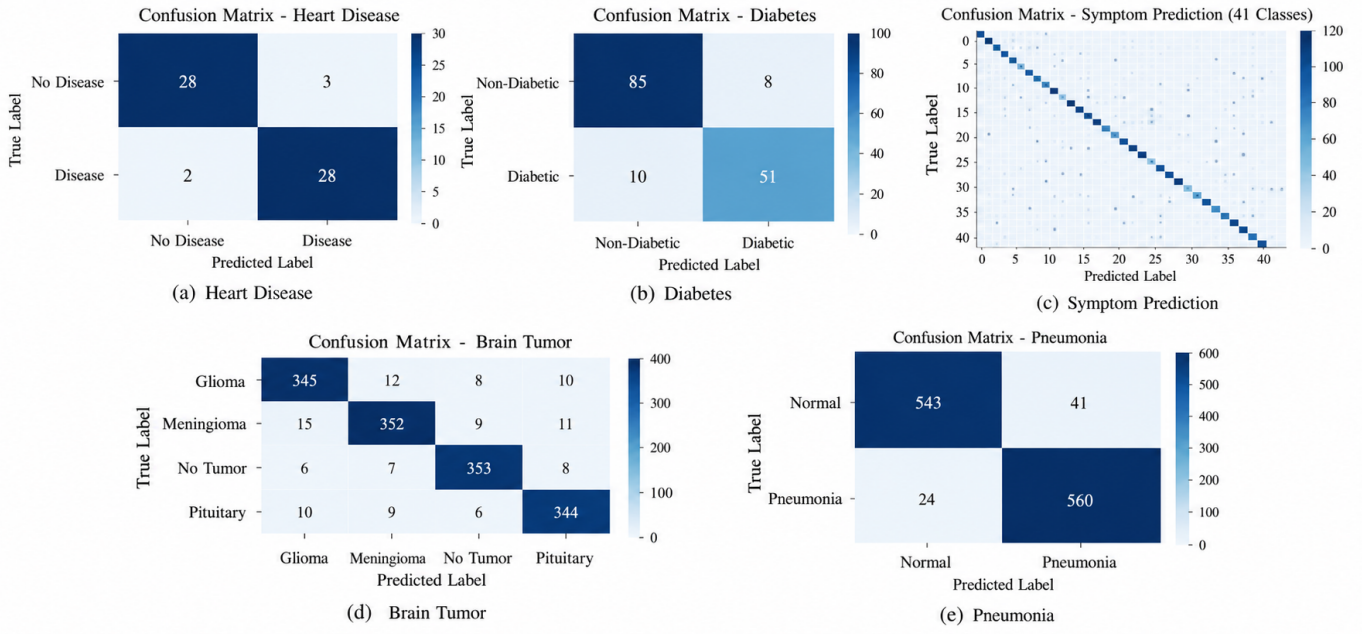


Fig. 3. Confusion matrices of the five disease-specific prediction modules on their respective test sets.

TABLE IV
COMPARISON OF EXISTING AI-BASED HEALTHCARE MODELS WITH NEUROVISIONX

Existing Approach	**Application**	**Limitation**	**NeuroVisionX**
Heart disease DL [1]	Heart disease	Single disease	Multi-disease prediction, explainability, reporting, and history.
Diabetes model [2]	Diabetes	Single structured-data task	Integrated with other disease-specific modules.
Brain tumor DL [4]	MRI classification	Brain tumor only	MRI diagnosis with Grad-CAM and automated reporting.
Pneumonia CNN [11]	X-ray classification	Single imaging task	Pneumonia detection with explainability and patient history.
Explainable thyroid model [5]	Disease prediction	Disease-specific	Explainability within a multi-disease platform.

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